Section B

Subject: Mathematics

- **Answer the following questions in a separate sheet and attach them to the worksheet.**

1. Which of the following $\frac{3}{7}$ and $\frac{5}{51}$ is greater.

2. Verify the associative law for the three rational numbers $\frac{1}{5}, \frac{3}{6}$ and $\frac{2}{3}$.

3. Simplify: $\frac{1}{5} + \frac{3}{10} - \frac{1}{2}$.

4. Divide the sum of $-\frac{7}{6}$ and $\frac{4}{5}$ by their product.

5. What rational numbers should be added to $-\frac{5}{8}$ so as to get $\frac{5}{9}$.

6. Leena walked $\frac{3}{4} km$ on Tuesday and she walked $1\frac{5}{6}$ times that distance on Wednesday. How far did she walk on Wednesday?

7. A shirt can be stitched using $2\frac{1}{4}$ m of cloth. How many shirts can be stitched using $31\frac{1}{2}$ m cloth?

8. The length and breadth of a rectangular park are $45\frac{1}{2}$ m and $34\frac{3}{4}$ m respectively. Find the perimeter and the area of the park.

9. A cyclist covers $14\frac{2}{5}$ km in $2\frac{1}{4}$ hours. What is the speed of the cyclist?

10. Fill in the blanks to make the statements true.

a. The equivalent of $\frac{5}{7}$, whose numerator is 45 is _____.

b. The reciprocal of a positive rational number is _____.

c. Zero has _____ reciprocal.

d. The negative of a negative rational number is always a _____ rational number.

e. The multiplicative inverse of $\frac{4}{3}$ is _____.

f. $\frac{1}{5} \times \left[\frac{2}{7} + \frac{3}{8}\right] = \left[\frac{1}{5} \times \frac{2}{7}\right] +$ _____.

11. A ribbon is $\frac{7}{8}$ meter long. It is cut into 4 equal pieces. What is the length of each piece?

12. Is 0 a rational number? Explain why?

13. What is the multiplicative identity of rational numbers?

14. A rational number is such that when added to $\frac{3}{5}$, the result is 0. Find the rational number.

15. Express 21^2 as the sum of two consecutive integers.

16. Is 324 a perfect square? If so, find the number whose square is 324.

17. The following numbers are not perfect squares. Give reason:

a. 1057 b. 23453 c. 222000

18. Find the smallest number by which 180 must be multiplied so that the product is a perfect square.

19. Without adding find the sum: $1 + 3 + 5 + 7 + 9 + 11 + 13 + 15 + 17 + 19$.

20. Express 121 as the sum of 11 odd numbers.

21. Write a Pythagorean triplet whose one member is 18.

22. Find the smallest square number that is divisible by each of the numbers 8, 15 and 20.

23. 1089 students are sitting in a cinema hall in such a manner that there are as many in a row as there are rows in the cinema hall. How many rows are there in the cinema hall?

24. Find the square roots of the following numbers by division method:

a. 4489 b. 5776 c. 24586

25. Find the least number which must be subtracted from 1989 so as to get a perfect square. Also, find the square root of the perfect square so obtained.